

HEAT KING



IDF1000 Operation Manual



As a new customer of Heat King we would like to welcome you! We are looking forward to providing you with technical support for your Heat King unit. Whatever you need we are here to help.

Ways to contact us for support

By Phone : 1-800-661-0304

The main Technical support phone line is staffed monday-friday 7:00 AM to 3:30 PM CST excluding Manitoba statutory holidays.

After hours support

Calls received outside of regular hours are directed to the On-call technician. After hours support is reserved for issues that cannot wait until the next business day for resolution. If no answer please leave a message and we will get back to you as soon as possible.

By Email : Support@tamarack-ind.com

Feel free to email us at any time with technical questions or parts inquiries. Please include the year make and model of your Heat King if you have a specific question about your machine so we can better help you. If it is an emergency please call 1-800-661-0304

For more information please visit our website
www.heat-king.ca



WARNING



CALIFORNIA - Proposition 65 Warning

Engine exhaust and some of its constituents and some dust created by power sanding, sawing, grinding, drilling and other construction activities contains chemicals known to the State of California to cause cancer, birth defects and other reproductive harm.

Some examples of these chemicals are:-

- Lead from lead-based paints
- Crystalline silica from bricks
- Cement and other masonry products
- Arsenic and chromium from chemically treated lumber

Your risk from these exposures varies, depending on how often you do this type of work. To reduce your exposure to these chemicals:

ALWAYS work in a well ventilated area, and work with approved safety equipment, such as dust masks that are specially designed to filter out microscopic particles.

Table of contents

Section		Title	Pages
	1	Warning	1
	2	Safety	3-9
		Introduction	10
	3	Training sign off	11
	4	Warranty	12-13
	5	How it works	14-15
		Controls	16-20
		Operating procedures	21
	6	Maintenance	22-29
	7	Service parts	30
	8	Troubleshooting	31-34
	9	Lifting instructions	35
	10	Specifications	36
	11	MSDS	37-40
	12	Schematics	41-44

SAFETY

SAFETY ALERT SYMBOL

This Safety Alert symbol means
ATTENTION! BECOME ALERT!
YOUR SAFETY IS INVOLVED!



The Safety Alert symbol identifies important safety messages on the Tamarack Heat King and in the manual. When you see this symbol, be alert to the possibility of personal injury or death. Follow the instructions in the safety message.

Why is SAFETY important to you?

3 Big Reasons

Accidents Disable and Kill
Accidents Cost
Accidents Can Be Avoided

SIGNAL WORDS:

Note the use of the signal words **DANGER**, **WARNING** and **CAUTION** with the safety messages. The appropriate signal word for each message has been selected using the following guide-lines:

DANGER - Indicates an imminently hazardous situation that, if not avoided, will result in death or serious injury. This signal word is to be limited to the most extreme situations, typically for machine components that, for functional purposes, cannot be guarded.

WARNING - Indicates a potentially hazardous situation that, if not avoided, could result in death or serious injury, and includes hazards that are exposed when guards are removed. It may also be used to alert against unsafe practices.

CAUTION - Indicates a potentially hazardous situation that, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.

Safety, Installation & Operation

SAFETY

YOU are responsible for the **SAFE** operation and maintenance of your Tamarack Industries Heat King. **YOU** must ensure that you and anyone else who is going to operate, maintain or work around the Heat King be familiar with the operating and maintenance procedures and related **SAFETY** information contained in this manual. This manual will take you step-by-step through your working day and alerts you to all good safety practices that should be adhered to while operating the Heater.

Remember, **YOU** are the key to safety. Good safety practices not only protect you but also the people around you. Make these practices a working part of your safety program. Be certain that **EVERYONE** operating this equipment is familiar with the recommended operating and maintenance procedures and follows all the safety precautions. Most accidents can be prevented. Do not risk injury or death by ignoring good safety practices.

- Heater owners must give operating instructions to operators or employees before allowing them to operate the machine, and at least annually thereafter.
- The most important safety device on this equipment is a SAFE operator. It is the operator's responsibility to read and understand ALL Safety and Operating instructions in the manual and to follow these. Most accidents can be avoided.
- A person who has not read and understood all operating and safety instructions is not qualified to operate the machine. An untrained operator exposes himself and bystanders to possible serious injury or death.
- Do not modify the equipment in any way. Unauthorized modification may impair the function and/or safety and could affect the life of the equipment.
- Think SAFETY! Work SAFELY!

GENERAL SAFETY

1. Read and understand the Operator's manual and all safety signs before operating, maintaining, adjusting, servicing or cleaning the Heater.
2. Only trained competent persons shall operate the Heater. An untrained operator is not qualified to operate the machine.



3. Have a first-aid kit available for use, should the need arise and know how to use it.



4. Do not allow riders.

5. Have a fire extinguisher available for use should the need arise and know how to use it.



6. Wear appropriate protective gear. This list includes, but is not limited to:

- A hard hat
- Protective boots with slip resistant soles
- Protective goggles
- Heavy gloves
- Hearing protection



7. Place all controls in their OFF position, disconnect power cords and wait for all moving parts to stop before servicing, adjusting or maintaining

8. Wear appropriate hearing protection when operating for long periods of time.



9. Wear protective gloves



10. Ventilation ~ Never operate in a poorly ventilated or enclosed area. Avoid prolonged breathing of exhaust gases.



11. Hot surface ~ Avoid contact with hot exhaust and glycol system. Allow to cool before performing repairs or service.



12. Electrocution Hazard ~ Always use proper size grounded extension cord. Inspect all extension cords for cuts, frayed wires and broken connectors. Do not use cords if not in good condition.



13. Fire Hazard ~ Do not operate machine in the vicinity of open flames, sparks or while smoking.



14. Explosion Hazard ~ Battery
Take care when handling battery (if installed)



ELECTRICAL SAFETY

1. Place all controls in their OFF position, disconnect power cords and wait for all moving parts to stop before servicing, adjusting or maintaining.
2. Place all controls in their OFF position before plugging in power cords.
3. Keep all electrical components in good repair before starting.
4. Do not lay power lines or connectors in water or on a wet surface. Dry connectors and raise power lines out of the water before and during operation.
5. Do not operate machine if there are electrical malfunctions. Correct problem before resuming work.

TIRE SAFETY

1. Failure to follow proper procedures when mounting a tire on a wheel or rim can produce an explosion which may result in serious injury or death.
2. Do not attempt to mount a tire unless you have the proper equipment and experience to do the job.
3. Have a qualified tire dealer or repair service perform required tire maintenance.
4. Torque wheel nuts to 165 ft-lbs

STORAGE SAFETY

1. Store unit in an area away from human activity.
2. Do not permit children to play on or around the stored Heater.



FUEL SAFETY



Danger: To avoid possible injury, fire, or explosion, please read and follow these instructions.

1. Handle fuel with care. It is highly flammable.
2. Allow burners to cool for 5 minutes before refueling. Clean up spilled fuel before restarting engine.
3. Do not refuel the machine while smoking or when near open flame or sparks.
4. Always use an approved fuel container.
5. Fill fuel tank outdoors.
6. Prevent fires by keeping machine clean of accumulated trash, grease and debris.

INSTALLATION SAFETY

1. Use only fuel for the heater burner specified in the **burner section of model specifications**. The use of incorrect fuel may result in fire or explosion and severe injury to the operator.

2. Fuel burning equipment must have proper ventilation for cooling, combustion air, and exhausting of combustion products

3. Personnel trained in and familiar with the type of equipment being serviced should only perform adjustments to fuel burning equipment.



WARNING: This machine emits carbon monoxide, a deadly gas, and must be vented if used in an enclosed area. Improper venting can cause poor combustion, delayed ignition, down drafts and possibility of freezing of the coil.

The information contained herein is offered for reference only. You must comply with local codes and investigate through your local gas or utility companies when installing, as there may be some local requirements you must comply with. Also see ANSI Z223.1

Never reduce the stack size, the diverter and stacking should be the same size as the stack opening on the machine

BEFORE INSTALLATION

1.LOCATION: This machine should be installed by only qualified technicians. The machine should be set upon a level surface, minimizing the effect of winds, rain and snow. Install the machine considering locations for fuel connections, electrical connections, venting, and maintenance. All wiring and electrical connections should comply with the National Electrical Code (NEC) and with local codes and practices.

2.GAS AND ELECTRICITY: Gas and electricity must be shut off when installing or servicing.

3.LOCAL CODES: Installation and servicing must only be performed by qualified personnel and must conform to local codes.

4.FIRE HAZARD: Keep combustible materials away from the machine. DO NOT allow lint or dust to collect in the burner area.

5. QUALIFIED PERSONNEL: All installation and servicing must only be performed by qualified personnel and must conform to the local codes.

MAINTENANCE SAFETY

1. Review the Operator's Manual and all safety items before working with, maintaining or operating the Heater.
2. Place all controls in their OFF position, disconnect power cords and wait for all moving parts to stop before servicing, adjusting or maintaining.
3. Follow good shop practices:
 - Keep service area clean and dry.
 - Be sure electrical outlets and tools are properly grounded.
 - Use adequate light for the job at hand.
4. Keep hands, feet, clothing and hair away from all moving and/or rotating parts.
5. Always wear heavy gloves to prevent burns when handling hot components. Wait until burners, coils and glycol system components have cooled before working on them.
6. Do not attempt any adjustment or maintenance to any system of the Heater unless the power wires are disconnected from the battery.
7. Make sure that all guards, shields and hoods are properly installed and secured before operating the Heater.
8. Securely support the machine using blocks or safety stands before working beneath it or changing tires.
9. Store and transfer diesel fuel, solvents, cleaners or any flammable liquids only in safety standard approved containers.



OPERATING SAFETY

1. Read and understand the Operator's Manual and all safety signs before operating, servicing, maintaining or adjusting the Heater.
2. Place all controls in their OFF position, disconnect power cords and wait for all moving parts to stop before servicing, adjusting or maintaining.
3. Do not allow riders in or on machine during transport.
4. Clear the area of bystanders, especially small children, before starting and operating.
5. Keep the working area clean and free of debris to prevent slipping or tripping. Clean up fuel spills immediately if they occur.
6. Slow down. Use care when working around unit - the steps, frame or floor may be wet and/or slippery.
7. Do not allow personnel that are taking drugs, alcohol or any medications that impair the senses or when excessively tired or stressed to operate the Heater.
8. Do not operate unit in a poorly ventilated or enclosed area to prevent asphyxiation when the heaters are operating.
9. Do not smoke when connecting fuel source or when working around machine.
10. Always wear heavy gloves when working on the machine to prevent burns when touching hot components.
11. Use the foot pedal switch to engage the hose reel take up or extend function and guide the hose by hand.
12. Keep all electrical lines and components in good working order. Do not operate in wet conditions or when standing in water. Damp or wet conditions can cause shocks or trip the breakers.
13. Keep all components in good condition.
14. Do Not plug or block access doors or vents. Keep 1 foot of clearance around unit.
15. Review safety instructions with operators annually.

SAFE TRANSPORTATION AND STORAGE



TRANSPORTING SAFETY

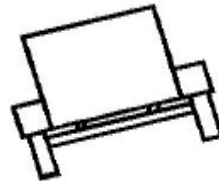
1. Attach to towing vehicle and secure with a mechanical retainer. Cross the safety chains under the hitch and anchor to truck frame.
2. Connect the brake anchor cable to the truck frame to activate the trailer brakes if the trailer unexpectedly unhooks. Provide sufficient slack for turning.
3. Check that all lights and reflectors required by the DOT are clean and functioning.
4. Do not exceed 55 mph under ideal conditions.
5. Do not allow riders on machine.
6. Do not drink and drive.

When transporting the machine, review and follow these instructions:

1. Be sure all bystanders are clear of the machine.
2. Back the truck up to the hitch and lower hitch over the ball.
3. Secure with a mechanical retainer.
4. Cross the safety chains under the hitch and attach to truck frame.
5. Attach the brake line to the truck frame. Be sure to leave sufficient slack for turning.
6. Connect electrical harness to truck plug-in.
7. Raise and secure the hitch jack.
8. Reverse the Heat King set-up procedure
9. Check and be sure that all lights are working.
10. Do not allow riders on machine.
11. Never exceed a safe travel speed.
12. Do not drink and drive.
13. Check with local highway authorities on the specific requirements for transporting fuel oil through their jurisdiction. Always comply with the requirements before transporting.



WARNING



TRAILER TOWING CAN BE HAZARDOUS

- * DO NOT exceed 55 mph under ideal conditions
- * Reduce speed under adverse weather, road or terrain conditions
- * Avoid sudden lane changes, U-turns etc.
- * Sudden maneuvers may cause tipping, rollover, jackknifing or sliding of the trailer and without warning loss of control of the towing vehicle may result.
- * Allow for increased braking distance due to weight of trailer
- * Read the Operator's Manual before towing.

STORAGE SAFETY

1. Store unit in an area away from human activity.
2. Do not permit children to play on or around the stored Heater.

PLACING IN STORAGE

After the season's use, the machine should be thoroughly inspected and prepared for storage. Repair or replace any worn or damaged components to prevent any unnecessary down time at the start of next season. To insure a long, trouble free life, this procedure should be followed when preparing the unit for storage:

1. Clear the area of bystanders, especially small children.
2. Thoroughly wash the machine using a pressure washer to remove all dirt, mud, debris and residue.
3. Inspect the hoses and burners for damage. Repair or replace damaged parts. Remove all entangled material.
4. Drain and remove the diesel fuel from the tank whenever the unit will be stored for more than 30 days. Removing the fuel during storage eliminates the formation of organic matter in the system and prevents leaks. Always refill the tank before its next use.
5. Lubricate all grease fittings. Make sure that all grease cavities have been filled with grease to remove any water residue from the washing.
6. Touch up all paint nicks and scratches to prevent rusting.
7. Move to storage area.
8. Select an area that is dry, level and free of debris.
9. Place blocks under the hitch if required.
10. Padlock each door.
11. Unhook from tow unit
12. If the machine cannot be placed inside, cover with a waterproof tarpaulin and tie securely in place.
13. Store the machine in an area away from human activity.
14. Do not allow children to play on or around the stored machine.

INTRODUCTION

Congratulations on your choice of a Tamarack Industries Heat King IDF1000 to complement your construction operation. This equipment has been designed and manufactured to meet the needs of the buyer for the efficient heating of construction sites.

Safe, efficient and trouble free operation of your IDF1000 requires that you and anyone else who will be operating or maintaining the Heater, read and understand the Safety, Operation, Maintenance and Trouble Shooting information contained in the Operator's Manual.



This manual is applicable to the IDF1000 built by Tamarack Industries. Use the Table of Contents as a guide when searching for specific information.

Keep this manual handy for frequent reference and to pass on to new operators or owners. Call your Heat King distributor or customer service if you need assistance or information at 1-800-661-0304

SIGN-OFF FORM

Tamarack Industries follows the general Safety Standards specified by the Society of Automotive Engineers (SAE) and the Occupational Safety and Health Administration (OSHA). Anyone who will be operating and/or maintaining the Heat King must read and clearly understand ALL Safety, Operating and Maintenance information presented in this manual.

Do not operate or allow anyone else to operate this equipment until such information has been reviewed. Annually review this information before the season start-up.

Make these periodic reviews of SAFETY and OPERATION a standard practice for all of your equipment. We feel that an untrained operator is unqualified to operate this machine.

A sign-off sheet is provided for your record keeping to show that all personnel who will be working with the equipment have read and understand the information in the Operator's Manual and have been instructed in the operation of the equipment.

SIGN-OFF FORM

[illegible]

TAMARACK INDUSTRIES

HEAT KING

WARRANTY REGISTRATION FORM & INSPECTION REPORT

WARRANTY REGISTRATION (please print)

This form must be filled out by the dealer and signed by both the dealer and the customer at the time of delivery.

Customer's Name _____

Distributor Name _____

Address _____

Address _____

City, State, Code _____

City, State, Code _____

Phone Number (____) _____

Check One:

Contact Name _____

Private ☐

Heat King Model _____

Contractor ☐

Serial Number _____

Other _ _ _ _

Delivery Date _____

DISTRIBUTOR INSPECTION REPORT

- ____ Tire Pressure Checked
- ____ Wheel Bolts Torqued
- ____ Brakes Work
- ____ Check Fluid Levels (Fuel and Glycol)
- ____ Lubricate Machine
- ____ Check That All Controls Function
- ____

SAFETY

- ____ Emergency Stop Switch Works
- ____ All Decals Installed and Legible
- ____ Lights and Reflectors Installed, Clean and Working
- ____ Review Operating And Safety Instructions

I have thoroughly instructed the buyer on the above described equipment which review included the Operator's Manual content, equipment care, adjustments, safe operation and applicable warranty policy.

Date _____

Dealer's Rep. Signature _____

The above equipment and Operator's Manual have been received by me and I have been thoroughly instructed as to care, adjustments, safe operation and applicable warranty policy.

Date _____

Owner's Signature _____

Tamarack Industries CONDITIONS OF SALES & LIMITED WARRANTY

All sales made by Tamarack Industries, here after referred to as Tamarack, a Division of ELJO Industries Inc. are subject to these conditions unless otherwise agreed in writing with a duly authorized officer of Tamarack. In all cases of conflict between these conditions and the requirements of the purchase order, these conditions shall prevail.

(1) SALES POLICY: Nothing herein shall be construed as abridging the right of Tamarack to sell directly or indirectly to: (a) Federal, State or Provincial Governments or Agencies thereof, or to Agencies employing Federal, State or Provincial Government aid; (b) Purchasers who buy Tamarack's products for sale as integral or assembled parts of their products; (c) Firms operating on a national scale; (d) Any other class of purchaser to whom Tamarack may from time to time, elect to sell.

(2) PRICES: All prices are F.O.B. our warehouses, freight allowance as specified on Distributor Net Price Lists. The suggested list prices and discounts schedules are established by Tamarack and are intended to act as a guide for our distributors. Unless otherwise stated in writing, prices are subject to change without notice and will be applied as in effect at time of shipment.

(3) TERMS: Unless otherwise agreed upon in writing by an officer of Tamarack, all invoices become due and payable net 30 days following the date in invoice. Interest at the maximum legal rate will be charged on all overdue accounts. Minimum net charge per invoice is \$75.00

(4) CANCELLATION AND CHANGES: No orders or sales may be cancelled or changed without the consent of Tamarack. At Tamarack's option cancelled/changed orders are subject to payment of cancellation charges equal to all costs incurred by Tamarack up to the date of cancellation/change.

(5) DELAYED DELIVERIES: Tamarack shall not be liable for any delay of merchandise for any cause whatsoever.

(6) CLAIMS: All goods shall be deemed delivered to purchaser at the time they are placed in the hands of carrier and consigned to purchaser: loss, damage or destruction of any said merchandise is assumed by purchaser. No claims may be made for shortages unless made in writing within ten days from receipt of merchandise.

(7) RETURN OF GOODS: Written permission from Tamarack must be obtained before returning any merchandise. All transportation charges must be borne by the purchaser. Credit for returned goods will be based on the original price paid, less 20%. Special parts or custom-built items cannot be returned for credit.

(8) LIMITATION OF LIABILITY: Tamarack's liability on any claim of any kind, including negligence, for any loss or damage arising out of, connected with, or resulting

from contract, or the performance or breach thereof, or the design, manufacture, sale, delivery, resale, installation, technical direction of installation, inspection, repair, operation or use of any equipment covered by or furnished under contract shall in no case exceed the price paid by the purchaser for the equipment. Tamarack also disclaims all purchaser for the equipment. Tamarack also disclaims all liability, whether in contract, tort, warranty, or otherwise, to any party other than purchaser.

(9) All Price Lists, Catalogues and other material shall remain the property of Tamarack and are subject to return on demand. The Suggested List Prices are established by Tamarack and are intended to act as a guide. All shipping weights shown are approximate.

LIMITED TAMARACK WARRANTY

For one year from date of purchase, Tamarack will replace or repair for the original purchaser, free of charge, any part or parts, found upon examination by any Tamarack Authorized Service Depot or by the Tamarack factory, to be defective in material or workmanship or both. Limitations: This warranty does not apply to any (i) hardware that has been damaged by abuse, accident, act of God, misuse, misapplication, improper installation, or shipment, and (ii) hardware that has been modified or has a serial number that has been removed or defaced. Customers may extend the Warranty Period through the purchase of an extended warranty. Equipment and accessories not manufactured by Tamarack are warranted only to the extent of the original manufacturer's warranty. All transportation charges on parts submitted for replacement or repair under this warranty must be borne by the purchaser. For warranty service contact your nearest Tamarack Authorized Service Depot.

Engine Warranty

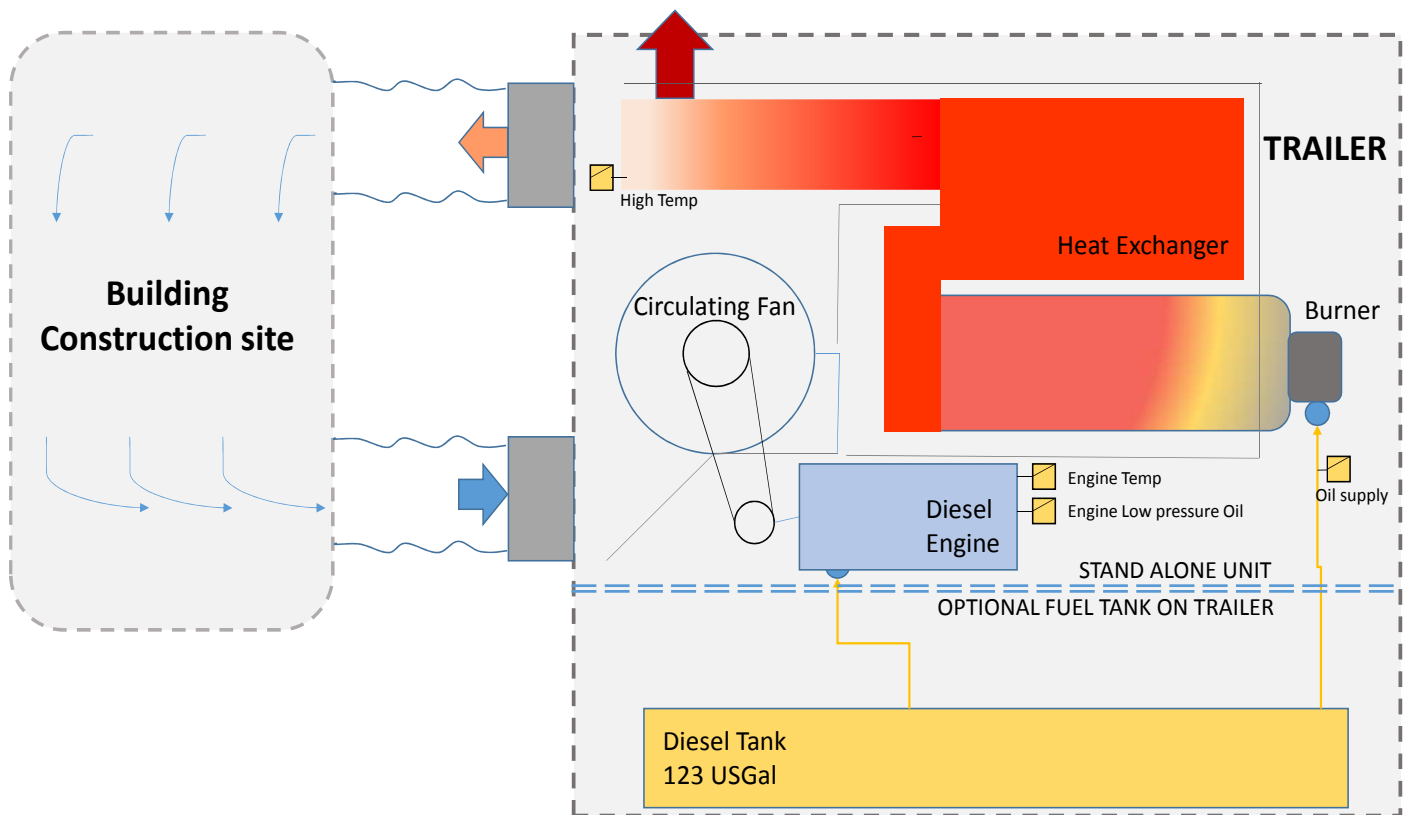
The engine warranty from Yanmar is 2 year/2000 hr. limited warranty; 3 year/3000 hr. Major component.

Optional extended warranty

The optional 2-year extended warranty is 2 years/ 3000 hours exclusive of the engine and subject to the same terms and conditions listed in the Limited warranty statement. Optional extended warranty must be purchased at time of original sale.

THERE IS NO OTHER EXPRESS WARRANTY, IMPLIED WARRANTIES, INCLUDING THOSE OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE LIMITED TO ONE YEAR FROM PURCHASE AND TO THE EXTENT PERMITTED BY LAW. LIABILITY FOR CONSEQUENTIAL DAMAGES UNDER ANY AND ALL WARRANTIES ARE EXCLUDED TO THE EXTENT EXCLUSION IS PERMITTED BY LAW. (THIS WARRANTY IS AN ADDITION TO ANY STATUTORY WARRANTY.)

How It Works



HOW THE MACHINE WORKS

Indirect fired:

This equipment was designed for space heating of buildings under construction, as an outdoor application with flue gases discharged outdoors. This unit does not require an external power supply.

A diesel operated engine is managing the fan via a centrifugal clutch. The fan drives ambient air through a large flue-air heat exchanger, and delivering the warm airstream to the building or construction site. The design also allows to recirculate the air from the building, while providing adjustable make up air.

A diesel fired burner provides the heating capacity. Optional, fuel is supplied by an external or an on-board diesel tank. A Programmable logic controller (PLC) manages the operation of the unit.

There are two alternatives to control the supply temperature of the unit: local or remote.

Local is a temperature sensor located at the end of the unit and monitors the temperature of the air leaving the unit. Remote is a thermostat located inside the site being heated, which requires a wire connection onto the unit. When the PLC detects the need for heat, (supply temperature below Set point), it powers up the burner while beginning to drive the engine's rpm up. As the rpm increases, the centrifugal clutch slowly engages the fan until the unit reaches the user-set fan speed.

Protective devices

The IDF1000 is equipped with the following protective devices to protect the user:

E-STOP

The emergency stop switch is monitored by the controller inputs. The input is connected to ground when the switch is pressed. In the event of an e-stop condition the system mode is set to OFF and all output functions are shut down by the controller.

MECHANICAL TEMPERATURE LIMITS

Two bi-metal thermostats are used in series to disable / lockout the burner and generate an alarm:

1. 87°C (190°F) Auto-Reset Type
2. 110°C (230°F) Manual Reset Type

All thermostats are physically mounted in the leaving air stream and wired to electrically disable the burner in the event of an over temperature condition. The controller monitors the status of these limits and generates a "Mechanical Temp Limit" alarm. This alarm condition will result in the unit being put into STANDBY mode. The operator would then be required to acknowledge the alarm condition and put the unit back into AUTO mode. Under normal operating conditions these alarms should not be triggered. Refer to page 33 for more information on alarms.

DIFFERENTIAL PRESSURE SWITCH

Used as feedback to verify that the centrifugal clutch is engaged and the Fan is operating. The pressure switch contains a normally open contact that closes when pressure is present which indicates the fan is running. During AUTO mode operation, this signal is monitored and used to put the unit into STANDBY if the contact has not closed after 15 seconds. It will then remain in STANDBY until the alarm is acknowledged and the system put back into AUTO mode by the operator.

Controls



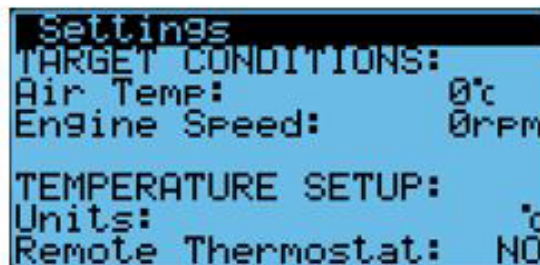
<i>Operator Panel Legend</i>	
1	Display
2	System Power Button
3	Engine Start / Stop Button
4	Heat Cycle On/Off Button
5	Temperature Up / Down Button
6	Air Speed Up / Down Button
7	System Power Indicator
8	Engine Status Indicator
9	Heat Cycle Status Indicator
10	Burner Flame Indicator
11	Fan Status Indicator

2. System Operation

1. **Power Up System** - Turn "System Power" switch to the ON position and wait for the controller to initialize. When ready the status message will change from "IO Setup, Please Wait..." to "Engine Stopped, Heat Cycle Off" (approximately 30 seconds):



2. **Verify Unit settings** – Use the [Up] / [Down] keys to scroll the Main menu and press the [Enter] key on the "Set" menu item. Use the [Enter] to move the cursor between fields. With the cursor on the Units field use the [Up] and [Down] key to toggle between C & F. Specify if control is to be based on a remote thermostat:



Note: If remote thermostat is set to NO, the unit will control the call for heat based on the supply air temperature and target defined by pressing the "Temperature Up" / "Temperature Down" operator panel buttons. If yes is selected, the call for heat is determined by dry contacts connected to the remote thermostat receptacle on the front of the unit.

3. **Start The Engine** - Press and hold the "Start Engine" button. The Engine status LED will flash indicating that the engine is being pre-heated (engine glow plugs turned on) prior to starting and the duration of pre-heat is based on ambient temperature, see table below. Once the pre-heat cycle is complete, starter will engage. Release "Start Engine" button when engine is running.

Pre-heat Duration vs. Outdoor Ambient Temperature	
Above 15C	2 Seconds
15C to 0C	5 Seconds
0C to -20C	10 Seconds
Below -20	15 Seconds

Note: When starting the engine in ambient conditions at and below -10C (14F), an engine warmup function will take place. During this period the status message at the bottom left corner of the main screen will indicate "Engine Warm Up" and any heat cycle request will be delayed until the warm up is complete. Once the warm up function is complete the status message will

change to "Engine Running". The length of time is determined by the ambient temperature; at -10C (14F) 5 minutes warm up is required and an additional 12 Seconds is added for every degree below that point. For conditions above -10C (14F) no warm up period is required.

2. **Adjust Air Temperature** – If a remote thermostat is not being used the "Temperature Up" / "Temperature Down" button to adjust the target supply air temperature (this can also be done on the fly at any point while the unit is in operation).
3. **Start Heat Cycle** - With the status message displaying "Engine Running, Heat Cycle OFF", turn the "Heat Cycle" switch to the ON position. At the beginning of a heat cycle the burner motor will cycle on and off 5 times to circulate fuel, during this period the status message will indicate "Warming Fuel". One complete the status message will change to "Heating..." and following an approximate 10 second heat exchanger air purge delay, the burner gas valve will open and ignition will take place.

***Note:** When the heat cycle is first initialized, the supply air fan will be delayed if the heat exchanger temperature is below 60C, this prevents blowing excessive cold air on start-up.*

4. **Adjust Air Speed** – With the supply air fan engaged, use the "Air Speed Up" / "Air Speed Down" buttons adjust the supply air CFM. The current supply air speed setting is displayed as a bar graph, along with the corresponding engine RPM on the main start-up screen:



5. Display Navigation

START-UP SCREEN



BUTTONS AND NAVIGATION



Alarm Key – Pressing this key will display any active alarms and provide read access to historical alarm data. When the key is flashing red, an active unacknowledged alarm is present. Solid red indicates an active acknowledged alarm.

```
Data logger Record:01
AL*04 13:33 01/11/16

Engine Over Temp!

Event:          Start
Air Temp:       0.00
Set Point:      24.00
```

```
Alarms

Press ALARM for 3s
to reset all alarms

Press ENTER
to DATA LOGGER
```



Program Key – This key provides access to the various settings of the IDF controller. A login password is required to make changes in this area.

```
Login

Insert Password: 0000
```



Exit Key – This key will take the operator back one screen. In any event it can be pressed repeatedly to return to the main start-up screen.



Up Key – The up key is used to change menu selections and adjust values within the various screens.



Enter Key – The enter key is used to select menu items, save values, and move the cursor between fields on the screen.



Down Key – The up key is used to change menu selections and adjust values within the various screens.

MAIN MENU

The main menu is used to access key settings and information about the IDF.



Set (Settings) Menu Item– Allows the operator to access setup options including temperature units (C or F) and change



i (information) Menu Item– The enter key is used to select menu items, save values, and move the cursor between fields on the screen.



I/O (inputs / outputs) Menu Item– The up key is used to change menu selections and adjust values within the various screens.

Operating Procedures

BEFORE STARTING THE UNIT

- * Ensure there is at least 2 feet (0.7 Meters) of clearance around machine.
- * If outdoor temperature is below 14°F or -10°C plug in the engine heater for 2-3 hours before starting. It requires a 120 VAC / 60 hz power supply.
- * Verify engine oil level.

TO START THE UNIT

1. Turn the Battery Disconnect switch - located behind the main door - to the ON position.
2. Turn the “Power” Switch to the ON position and wait for the Home Screen to appear on the display.

3. Two settings are accessible to the user at the display:

- * Temperature unit: °C or °F (default)
- * Remote Thermostat: YES or NO (default)

To edit the default settings, from the main screen press  until “Set” appears at the bottom right hand corner of the display.

Press  twice to access to the Temperature unit. Use  to toggle between °F and °C.

Press  once to move the cursor to the Remote thermostat line. Use  to toggle between NO and YES.

Press Esc to return to Main screen.

4. Press and hold the “Engine” switch to the start position. There will be a short delay to prepare the engine for starting. (indicator light will flash during this time). The colder the ambient, the longer the delay. If the engine is cold, let it warm up a few minutes before next step.

5. Turn the “Heating Cycle” switch to the ON position. The machine will start now the controlled heating cycle.

6. Using the corresponding switches, adjust the “Supply Temperature” and “Fan speed” Settings as needed or leave at the default settings.

TO STOP THE UNIT

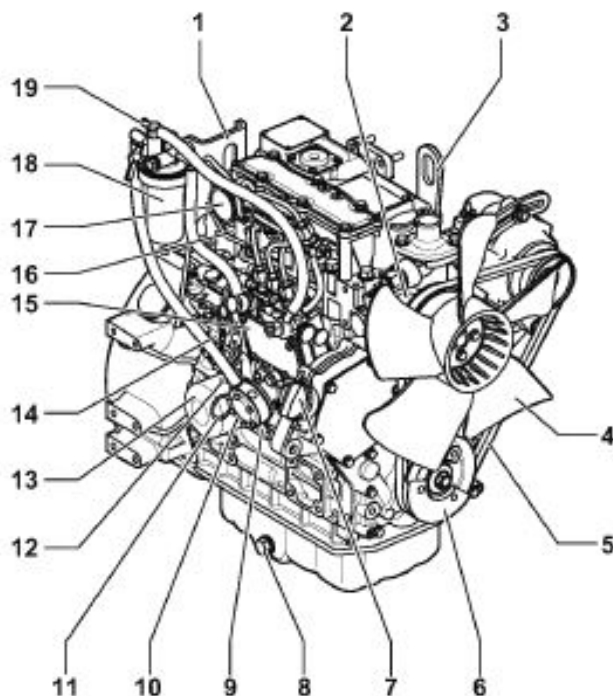
1. Turn the “Heating cycle” switch to the OFF position and wait for the Cool Down period to finish. (indicator light will stop flashing)
2. Press and hold the “Engine” switch to the Stop position until the engine shuts off.
3. Turn the “Power” Switch to the OFF position.

Maintenance

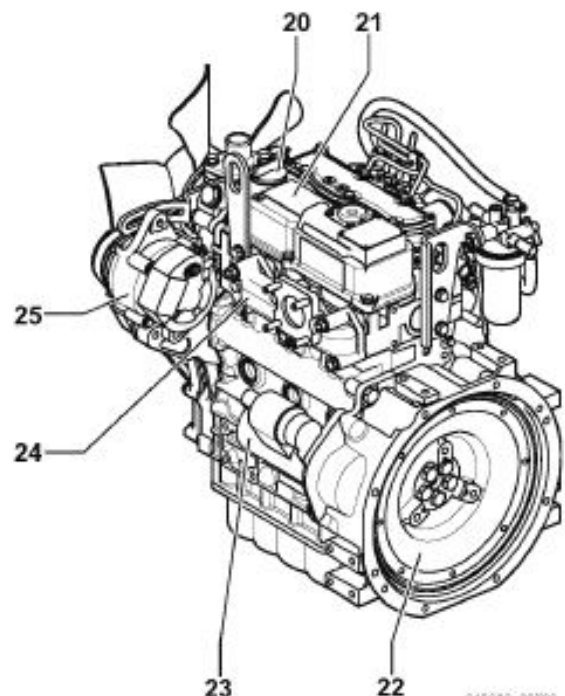
On the IDF1000 there are 4 areas of maintenance.

1. Engine
2. Burner
3. Blower belt
4. Heat Exchanger

Engine



- 1 – Lifting eye (flywheel end)
- 2 – Engine coolant pump
- 3 – Lifting eye (engine cooling fan end)
- 4 – Engine cooling fan
- 5 – V-belt
- 6 – Crankshaft V-pulley
- 7 – Side filler port (engine oil)
- 8 – Drain plug (engine oil)*
- 9 – Fuel inlet
- 10 – Mechanical fuel pump
- 11 – Dipstick (engine oil)
- 12 – Fuel priming lever
- 13 – Engine oil filter



- 14 – Governor lever
- 15 – Fuel injection pump
- 16 – Intake manifold
- 17 – Air intake port (from air cleaner)
- 18 – Fuel filter
- 19 – Fuel return to fuel tank
- 20 – Top filler port (engine oil)
- 21 – Rocker arm cover
- 22 – Flywheel
- 23 – Starter motor
- 24 – Exhaust manifold
- 25 – Alternator

045090-00000

Engine maintenance chart

Daily and periodic maintenance is important to keep the engine in good operating condition. The following is a summary of maintenance items by periodic intervals. Periodic maintenance intervals vary depending on application, loads and operating conditions. The following should be treated as a general guideline.

NOTICE. Establish a periodic maintenance plan according to the engine application and make sure you perform the required maintenance at the intervals indicated. Failure to follow these guidelines will impair the engines performance and shorten the engines life and may affect the warranty coverage on your engine. *

System	Check item	Daily	Periodic maintenance interval						
			Every 50 hours	Every 250 hours	Every 500 hours	Every 1000 hours	Every 1500 hours	Every 2000 hours	Every 3000 hours
Cooling system	Check and refill engine coolant	○							
	Check and clean radiator fins			○					
	Check and adjust cooling fan V-belt		○ 1st time	○ 2nd and after					
	Drain, flush and refill cooling system with new coolant					◇ or every 1 year whichever comes first			
Cylinder head	Adjust intake/exhaust valve clearance					●			
	Lap intake/exhaust valve seats (if required)							●	
Electrical equipment	Check indicators	○							
	Check battery		○						
Engine oil	Check engine oil level	○							
	Drain and fill engine oil		◇ 1st time	◇ 2nd and after					
	Replace engine oil filter								

* For more information regarding the engine please consult the Yanmar manual or contact yanmar directly.

Changing the engine oil

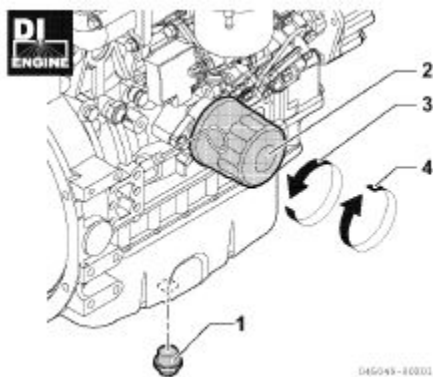
To avoid personal injury:

Be sure to stop the engine before draining engine oil.

When draining engine oil, place some container underneath the engine and dispose it according to local regulations.

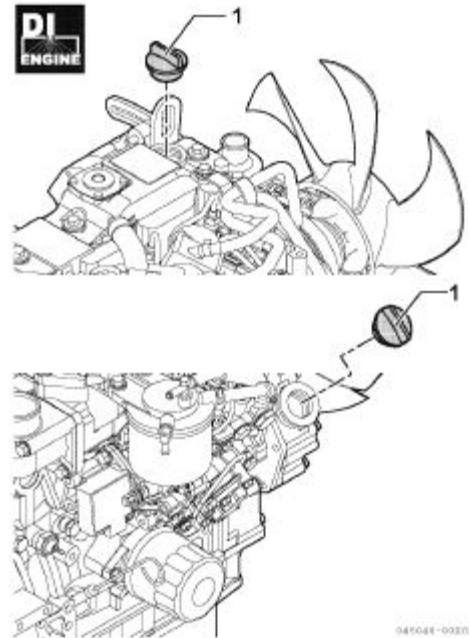
Do not drain oil after running the engine. Allow engine to cool down sufficiently.

1. Change oil after the initial 50 hours of operation and every 250 hours thereafter.
2. Remove the drain plug at the bottom of the engine(#1)and drain all the old oil. Drain oil easier and completely while the engine is warm.

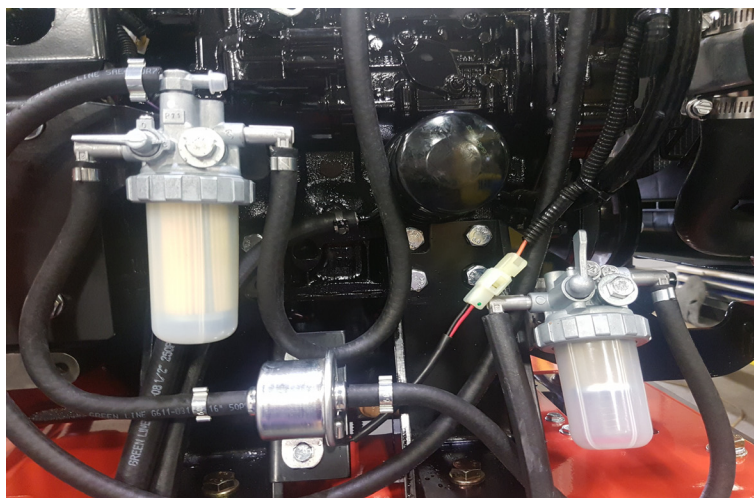


1. Replace the oil filter cartridge after the initial 50 hours of operation and every 250 hours thereafter.
2. Remove the old oil filter cartridge (#2) by turning counter clockwise (#3) with a filter wrench.
3. Clean the filter mounting surface and apply a film of oil to the gasket for the new cartridge.
4. Install the new engine oil filter manually by turning it clockwise (#4) When the gasket contacts the seal surface, tighten the cartridge one additional turn.

5. Add new oil to the engine through either of the oil filler ports (#1)



Fuel filter and water separator



Replace the fuel filter at specified intervals to prevent contaminants from adversely affecting the diesel fuel flow.

1. Stop the engine and allow it to cool.
2. Close the fuel shut off valve.
3. Loosen retaining ring.
4. Carefully remove the cup, pour the fuel into an approved container and dispose of waste properly.
5. Remove the fuel filter by pulling it down.
6. Replace the fuel filter with a new one (see pg.)
7. Wash the inside of the cup.
8. Check condition of the O-ring, replace if necessary.
9. Install the cup to the mounting flange and tighten the retaining ring, hand tighten only.
10. Open the fuel off valve.
11. Check for leaks.

Drain the fuel/Water separator whenever there are contaminants such as water collected in the bottom of the cup. Never wait until the scheduled periodic maintenance if contaminants are found.

The separator cup is transparent. In the cup is a red colored float ring. The float ring will rise to the surface of the water to show how much needs to be drained.

1. Position approved container under the fuel/water separator to collect the contaminants.
2. Close the fuel shut off valve.
3. Loosen the retaining ring.
4. Carefully remove the cup, retaining spring and float from the cup. Pour the contaminated fuel into an approved container and dispose of waste properly.
5. Clean the inside of the cup.
6. Inspect the condition of the mesh filter, clean as necessary.
7. Inspect the condition of the O-ring and replace as necessary.
8. Put the float and retaining ring back inside the cup.
9. Reinstall the cup to the mounting flange and tighten the retaining ring, hand tighten only.
10. Open the fuel shut off valve.
11. Check for fuel leaks.

Air filter



The engine performance is adversely affected when the air cleaner element is clogged with dust. Be sure to clean the air filter element periodically.

NOTICE: When the engine is operated in dusty conditions clean the air filter element more frequently. Never operate the engine with the air filter element removed. This may allow foreign material to enter the engine and cause damage.

1. Unlatch and remove the air cleaner cover.
2. Remove the filter element.
3. Replace the filter element with a new one if the element is damaged, excessively dirty or oily.
4. Clean the inside of the air cleaner cover.
5. Reinstall the air cleaner cover making sure you match the arrow on the cover with the arrow on the case.
6. Latch the air cleaner cover to the case.

Burner

CHANGING FUEL NOZZLES AND ELECTRODES

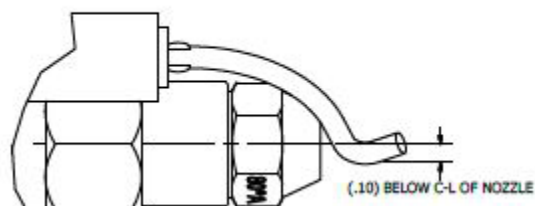
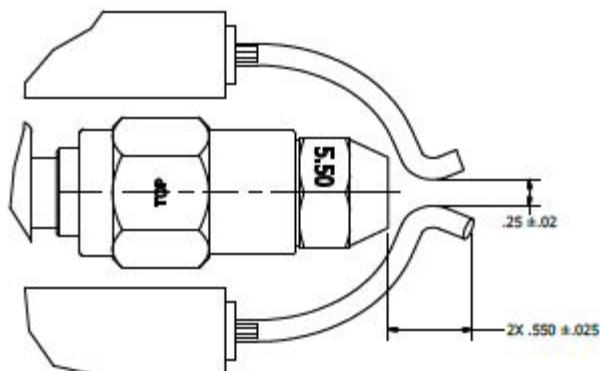
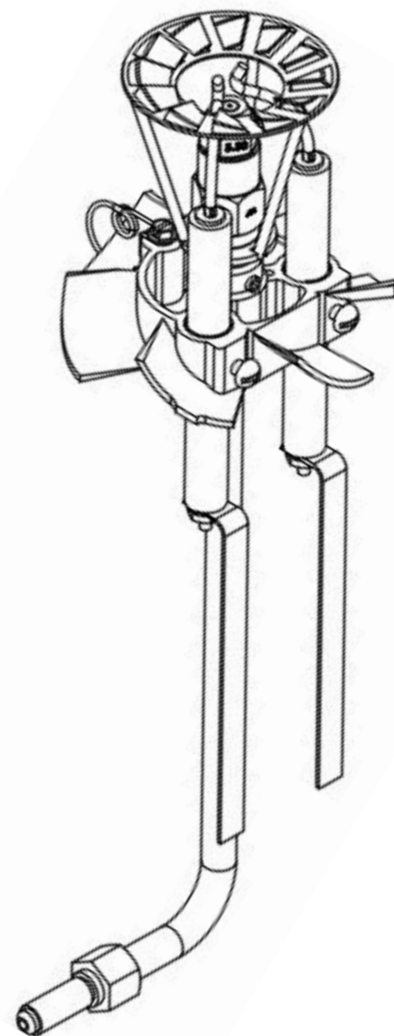
Annually or every 750 hours (or more often if fuel quality is poor) the fuel nozzels and elecrodos should be changed. To change, follow this procedure:

1. Remove the drawer assembly (fuel tube, flame turbulator, fuel nozzle, electrodes) from the burner chasis. .
2. Loosen the set screws that hold the electrode in place and remove them. Note there is a left and a right side electrode.
3. Remove the nozzle with a standard wrench and also use a backup wrench to keep the drawer assembly base from turning.
4. Replace the nozzle with the appropriate size for your Heat King model.
NOTE: use semi solid or hollow cone spray pattern nozzels.

IDF1000 use a 5.5 GPH, 80° nozzle @ 125 psi

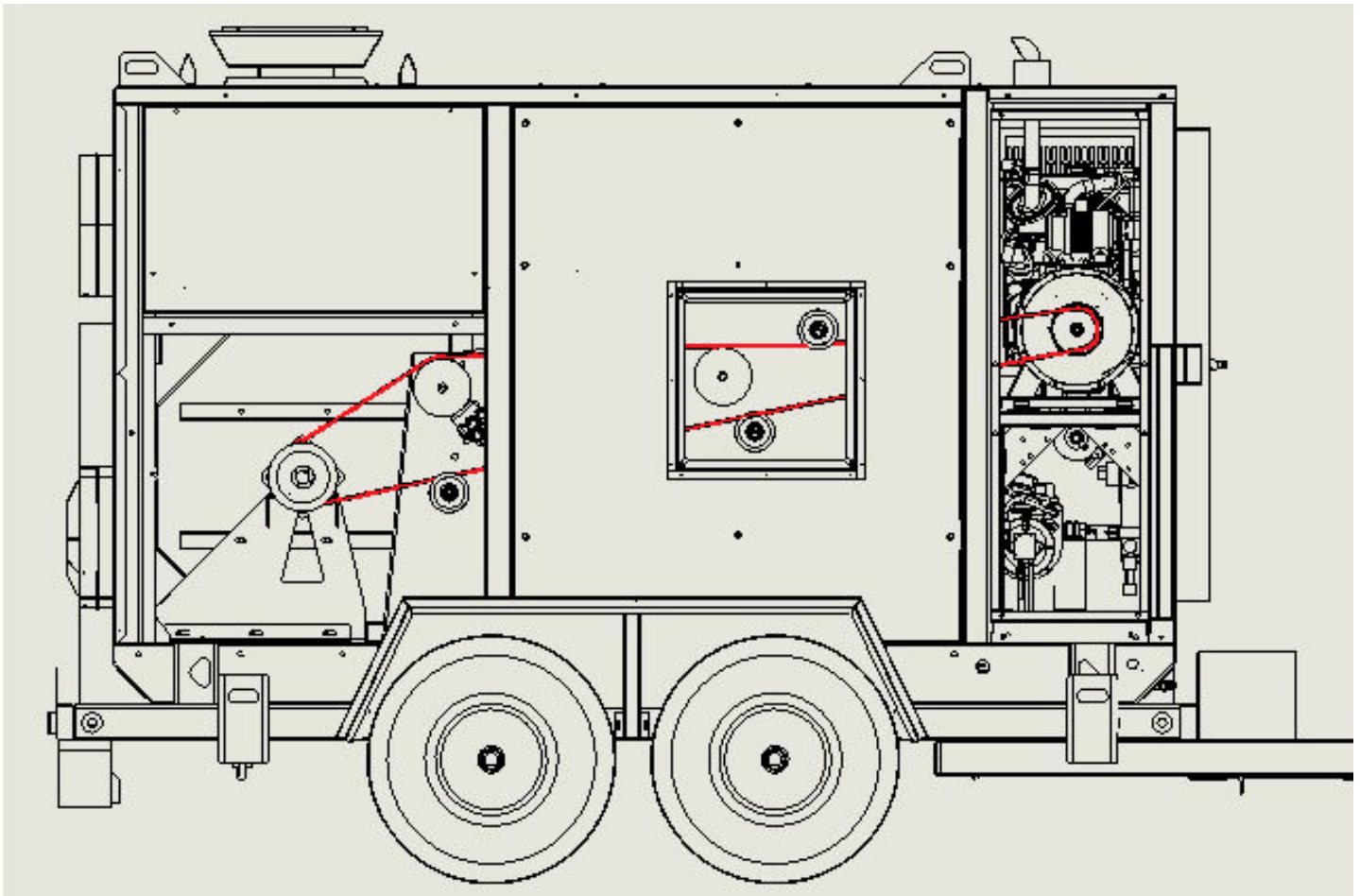
NOTE : FOR HIGH ALTITUDE USE (4000+ ft above sea level). Use the appropriate size for your heat king model.

5. Make sure nozzle is tight and then install the electrodes, set the gap and hieght according to this diagram, or use the appropriate wayne burner gap gauge.



Blower Belt

Belt Replacement



Replacing the blower drive belts:

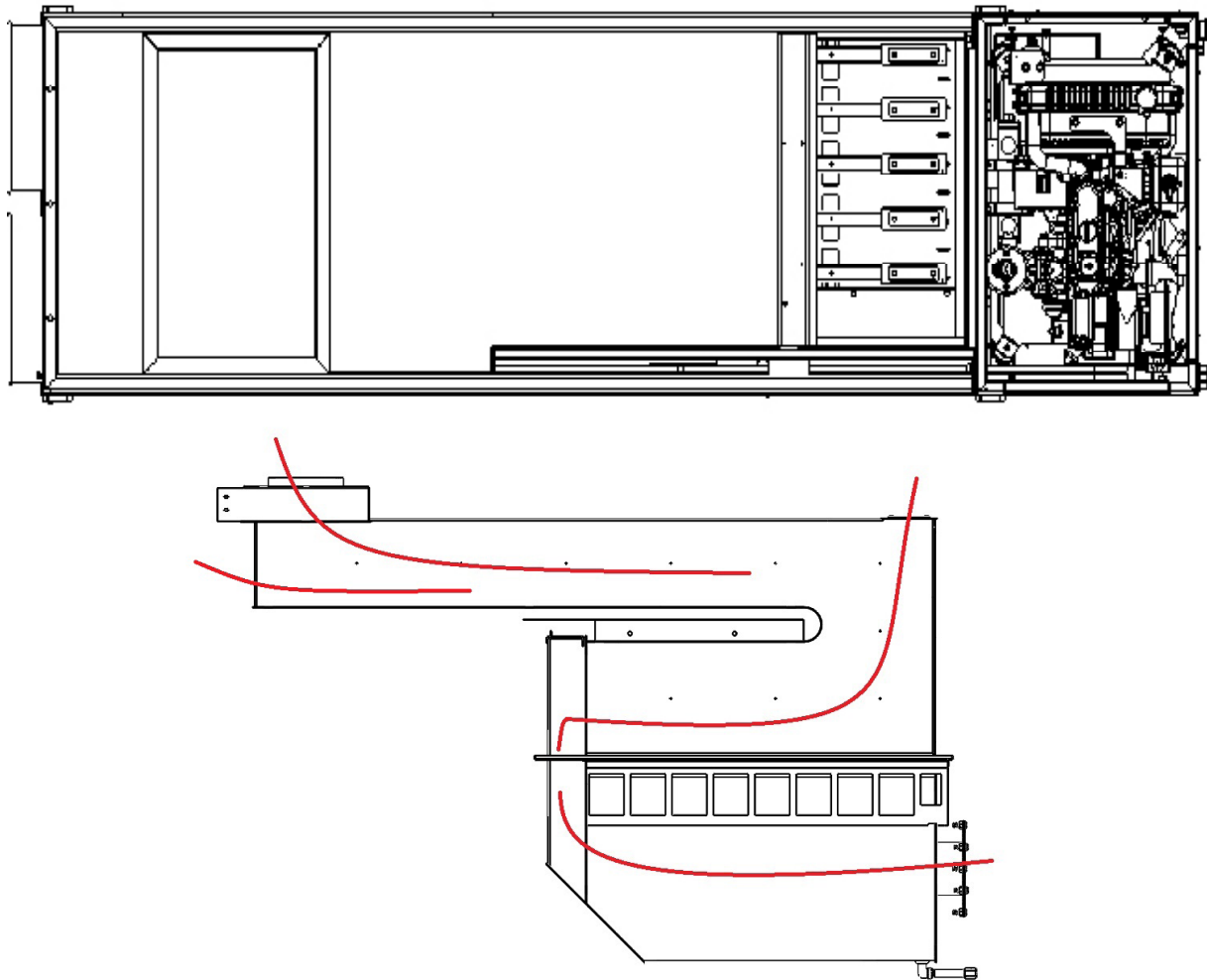
Tools required: Phillips driver, 3/4" wrench (replacement belt part # 460693)

- Remove 2 inspection doors on the right side of the unit using a Phillips drive.
- Using a 3/4" wrench, push down on the tensioner pulley to remove tension from the belts
- Remove outer belt first then inner belt
- Feed new belts through the belt channel from the engine side, ensure they are in the proper pulley grooves
- Using a 3/4" wrench, push down on the tensioner pulley
- Install inner belt on the tensioner then the outer,
- Release tension
- Install all 2 inspection doors

Heat Exchanger

Heat Exchanger Cleanouts

Inspect Heat Exchanger annually and clean as necessary using cleanout brushes and heat exchanger cleaning chemicals.



Cleaning the Heat Exchanger

Tools Required: 3/8" wrench, 7/16" wrench, 5/8" wrench, shop vac, cleanout brushes, chemicals as required.

- Remove inspection panel above engine using Phillips drive
- Remove Cleanout access covers using 7/16" wrench
- Remove exhaust chimney using 3/8" wrench
- Remove Burner assembly using 5/8" wrench
- Using Cleaning brush, clean each fin of the heat exchanger through the cleanout ports and exhaust opening
- Clean combustion chamber through burner opening, use shop vac to remove loose soot.

Service Parts

Item	Part #
Burner	
5.5 GPH 80°A Nozzle	460708
Left Electrode	461055
Right Electrode	461055
Cad cell	461057
Transformer	461053
Gap Guage Tool	461017
Primary Control	461056
Fuel Filter	445157
Burner Gasket	460408
Engine	
Oil Filter	461555
Fuel Filter Element	461556
Air Filter	461557
Water separator	461558
Alternator Belt	461477
Water Pump Belt	461559
Electric Fuel Pump	461560
Blower	
Drive Belt	460693
Clutch assembly	460480
Pillow Block Bearings	460749
High Limit switch (auto)	460434
High Limit Switch (manual)	460436
Trailer	
Tail Light (RED)	477009
Marker Light (Amber)	477007
Marker Light (red)	477008

Engine Troubleshooting

■ When it is difficult to start the engine

Cause	Countermeasures
Fuel is thick and doesn't flow.	* Check the fuel tank and fuel filter. * Remove water, dirt and other impurities. * As all fuel will be filtered by the filter, if there should be water or other foreign matters on the filter, clean the filter with kerosene.
Air or water mixed in fuel system	* If air is in the fuel filter or injection lines, the fuel pump will not work properly. * To attain proper fuel injection pressure, check carefully for loosened fuel line coupling, loose cap nut, etc. * Loosen air vent screws stop fuel filter and fuel injection pump to eliminate all the air in the fuel system.
Engine oil becomes thick in cold weather and engine cranks slow.	* Change grade of oil according to the weather (temperature.)
Battery is discharged and the engine will not crank.	* Charge battery. * In winter, always remove battery from machine, charge fully and keep indoors. Install in machine at time of use.

■ When output is insufficient

Cause	Countermeasures
Fuel is insufficient.	* Check fuel system.
Overheating of moving parts	* Check lubricating oil system. * Check to see if lubricating oil filter is working properly. * Filter element deposited with impurities would cause poor lubrication. Change element.
Air cleaner is dirty	* Clean the element every 100 hours of operation.
Injection pump wear	* Do not use poor quality fuel as it will cause wear of the pump. Only use No. 2-D diesel fuel. (See "FUEL" in "PERIODIC SERVICE" section.)

■ When engine suddenly stops

Cause	Countermeasures
Lack of fuel	* Check the fuel tank and refill the fuel, if necessary. * Also check the fuel system for air or leaks.
Bad nozzle	* If necessary, replace with a new nozzle.
Moving parts are overheated due to shortage of lubrication oil or improper lubrication.	* Check amount of engine oil with oil level gauge. * Check lubricating oil system. * At every 2 times of oil change, oil filter cartridge should be replaced.

■ When color of exhaust is especially bad

Cause	Countermeasures
Fuel is of extremely poor quality.	* Select good quality fuel. - Use No. 2-D diesel fuel only.
Nozzle is bad.	* If necessary, replace with new nozzle.

■ When engine must be stopped immediately

Cause	Countermeasures
Color of exhaust suddenly turns dark.	* Check the fuel injection system, especially the fuel injection nozzle.
Bearing parts are overheated.	* Check the lubricating system.
Oil lamp lights up during operation.	* Check the lubricating system. * Check the function of the relieve valve in the lubricating system. * Check pressure switch. * Check filter base gasket.

Troubleshooting

The Tamarack Construction Heat King IDF1000 is a self-contained heating system that can be used maintain a work area at a constant temperature. It is a simple system that requires minimal maintenance.

In the following trouble shooting section, we have listed many of the problems, causes and solutions to the problems which you may encounter.

If you encounter a problem that is difficult to solve, even after having read through this trouble shooting section, please contact your authorized dealer, distributor or the factory. Before you call, please have this Operator's Manual and the serial number from your machine ready.

PROBLEM	CAUSE	SOLUTION
Burner won't start.	No fuel.	Fill the fuel tank.
	No power.	Inspect wiring, check alternator output Breaker tripped. Reset breaker.
	E-Stop pressed.	Pull out Emergency Stop switch.
Burner gives off black smoke.	Wrong fuel/air mixture.	Reset air mixture.
	Fouled burner.	Clean burner.
	Plugged fuel filter.	Change filter.
No Air Flow.	Broken drive belts.	Replace drive belts.
	Drive belt loose.	Adjust drive belt tension.
	Clutch not engaging	Inspect clutch

Alarms

When an active alarm is present, the [Alarm] key will flash red. Pressing the key will take the operator to the main alarm screen:



```

Alarms 01/01
AL*04 13:33 01/11/16

Engine Over Temp!

Air Temp: 0.00
Set Point: 24.00
    
```

ID	Description	Cause	Troubleshooting
1	Too many mem writings!	Internal PLC Error	Reset PLC Replace PLC if persists
2	Retain mem write error!	Internal PLC Error	Reset PLC Replace PLC if persists
3	Low Oil Pressure!	Loss of oil pressure	Check oil level Inspect for oil leaks
4	Engine Over Temp!	Engine over heating	Check coolant level Check belt
5	Alternator Fail!	Alternator not charging	Check alternator Check belt
6	Burner Flame Loss!	Burner flame not detected	Check cad cell Check burner primary for errors Check battery voltage (>12.5VDC)
7	Air Over Temp!	Supply air exceeded 85C	Check drive belt Check clutch Check throttle actuator
8	Fan Failure!	Loss of air flow	Check Drive belt Check clutch Check pressure switch Check throttle actuator
9	Air Temp Switch!	Supply air limit exceeded	Check drive belt Check clutch Check throttle actuator Manually reset air limit
10	pCOe COMs Error!	PLC communication error	Reset PLC wait at least 30 seconds
11	pCOe Config Error!	PLC communication error	Reset PLC wait at least 30 seconds
12	Service Engine!	Engine runtime exceeded service interval	Service engine
13	Service Burner!	Burner runtime exceeded service interval	Service burner
14	Estop Pressed!	Emergency stop button pressed	Turn emergency stop button to reset.

ALARM RESET

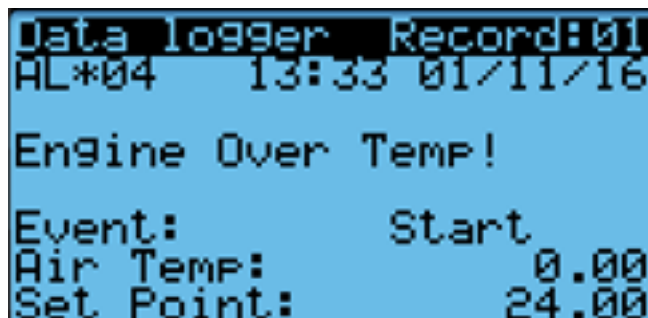
Alarms can be reset by the operator through the alarm screen. This is done by pressing and holding the [Alarm] for 3 seconds:



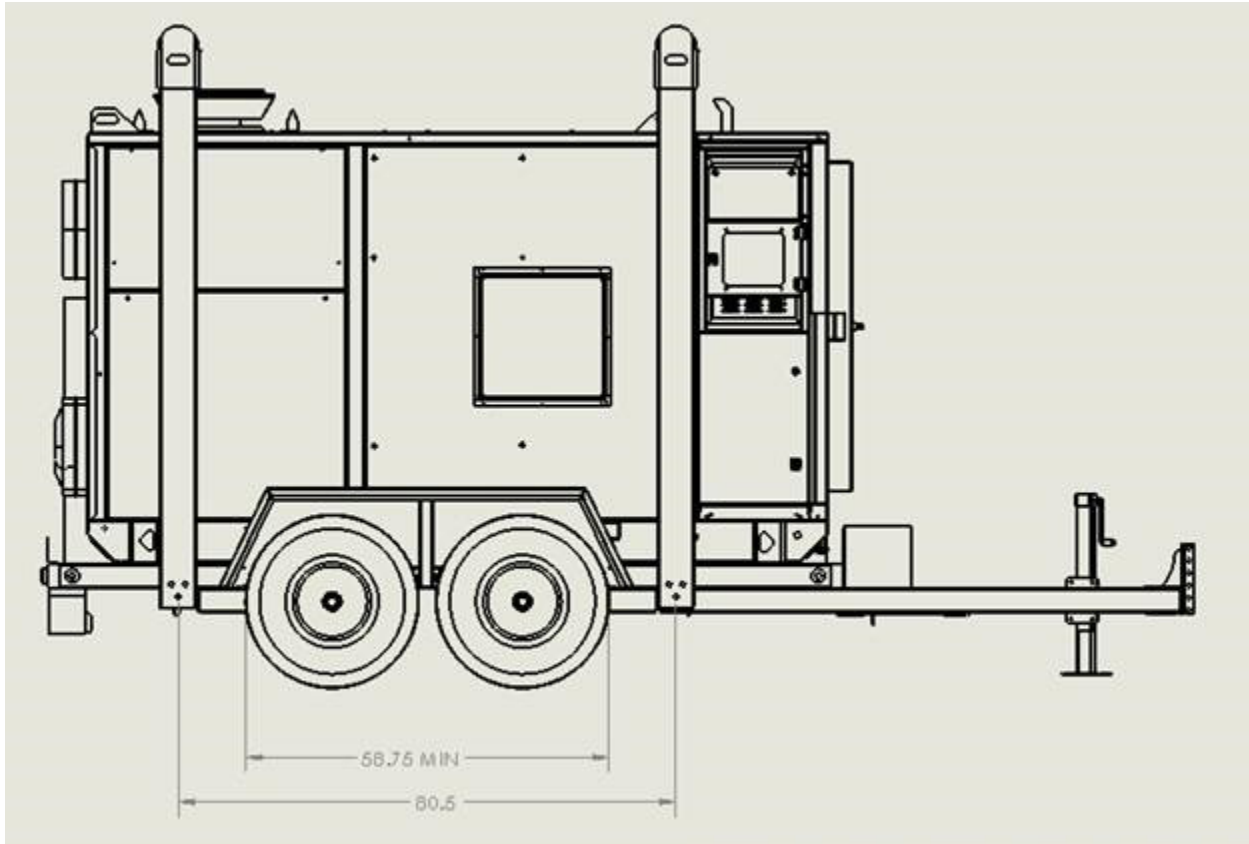
ALARM LOGGER

The alarm logger contains a list of historical alarm events for the purpose of troubleshooting. The logger can be accessed through the alarm screen by pressing the [Enter] key:

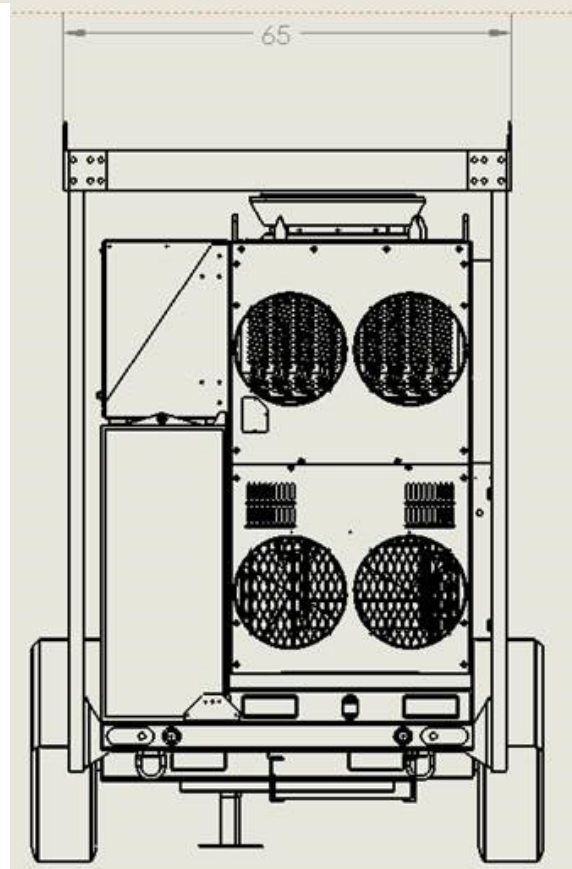
The records can be scrolled through using the up and down keys.



Lifting Instructions



The IDF1000 can be lifted by forklift with the forklift pockets located on either side of trailer tires or with the optional lifting frame.



IDF1000 Specifications

General Capacities and Component Specifications

Height (w/ heater vent)	77 in.
Width	55 in.
Length	129 in..
Weight	3360 lbs.
Engine	Yanmar Tier 4 diesel
Burner	Wayne MSR-DC
Clutch	Centrifugal 1600 rpm engagement

Optional Trailer With Fuel tank

Height (W/heater vent)	96 in.
Width	74 in.
Length	187 in.
Weight (Fuel Empty/Full)	5644/7990 Lbs
Axles	2 - 5000 lb. rated w/ Electric Brakes
Tires	ST225/75R-15
Tire inflation pressure	50 psi.
Torque of wheel nuts	165 ft-lbs
Hitch	2-5/16" Ball, alternatively Pintle
Tie Downs	4 - For Transporting
Fuel tank	326 US gallons
Ground Clearance	13 in.

Performance Specifications

Nominal Heat Input	1,000,000 BTU/h
Fuel Requirement	Low Sulfur Diesel
Run Time (with optional fuel tank)	Min 40Hrs @ 100%
Supply Air Ducts	2 x 16"(1x 24" or 1x20" option)
Return Air Ducts	2 x 16"(1x 24" or 1x20" option)
Air Flow	Max 9000CFM @ 5" W.C.
Fuel Consumption	7.8 GPH Max



WHMIS (Pictograms)	WHMIS (Classification)	Protective Clothing	TDG (pictograms)
 	B-3, D-2B	  	

Section 1. Chemical Product and Company Identification

Product Name	DIESEL FUEL	Code	W104 SAP: 120, 121, 122, 287
Synonym	Diesel 50, Diesel 50 LS, #1 Diesel, #1 Diesel LS, Diesel LC, Seasonal Diesel, Seasonal Diesel LS, Diesel AA, Domestic Marine Diesel, International marine Diesel, Seasonal Diesel Locomotive, Domestic Marine diesel LS, diesel -20°C (LS), LSD, Low Sulphur Diesel, dyed diesel, marked diesel, coloured diesel, Naval Distillate.	Validated on	3/2/2001.
Manufacturer	PETRO-CANADA P.O. Box 2844 Calgary, Alberta T2P 3E3	In case of Emergency	Petro-Canada: 403-296-3000 Canutec Transportation: 613-996-6666 Poison Control Centre: Consult local telephone directory for emergency number(s).
Material Uses	Diesel fuels are distillate fuels suitable for use in high and medium speed internal combustion engines of the compression ignition type.		

Section 2. Composition and Information on Ingredients

			Exposure Limits (ACGIH)		
Name	CAS #	% (V/V)	TLV-TWA(8 h)	STEL	CEILING
1) Diesel oil. 2) Proprietary additives. 3) Aromatic content is 50% maximum (benzene: nil). 4) * Notice of Intended Change (2000): 100 mg/m ³ , skin, A3.	68334-30-5 Not available	>99.9 <0.1	Not established* Not established	Not established Not established	Not established Not established
Manufacturer Recommendation	Not applicable				
Other Exposure Limits	Consult local, state, provincial or territory authorities for acceptable exposure limits.				

Section 3. Hazards Identification.

Potential Health Effects	Eye contact may cause mild eye irritation. Skin contact can cause moderate to severe irritation and produce drying, cracking, or defatting dermatitis. Inhalation of vapours can cause CNS depression with symptoms of nausea, headaches, vomiting, dizziness, fatigue, light-headedness, reduced coordination, unconsciousness and possibly death. Inhalation can also cause irritation of nose and throat. Aspiration of liquid drops into the lungs may produce potentially fatal chemical pneumonitis (fluid in the lungs), severe lung damage, or respiratory failure. For more information, refer to Section 11.
---------------------------------	--

Section 4. First Aid Measures

Eye Contact	IMMEDIATELY flush eyes with running water for at least 15 minutes, keeping eyelids open. Seek medical attention.
Skin Contact	Remove contaminated clothing - launder before reuse. Wash gently and thoroughly the contaminated skin with running water and non-abrasive soap. Seek medical attention.
Inhalation	Evacuate the victim to a safe area as soon as possible. If the victim is not breathing, perform artificial respiration. Allow the victim to rest in a well ventilated area. Seek medical attention.
Ingestion	DO NOT induce vomiting because of danger of aspirating liquid into lungs. Seek medical attention.
Note to Physician	Not available

Section 5. Fire-fighting Measures

Flammability	Class II - combustible liquid (NFPA).	Flammable Limits	LOWER: 0.7%, UPPER: 6%
Flash Points	Diesel Fuel: Closed Cup: >40°C (>104°F) Marine Diesel Fuel: Closed Cup: >60°C (>140°F)	Auto-Ignition Temperature	225°C (437°F)
Fire Hazards in Presence of Various Substances	Flammable in presence of open flames, sparks, or heat. Vapours are heavier than air and may travel considerable distance to sources of ignition and flash back. This product can accumulate static charge and ignite. May accumulate in confined spaces.	Explosion Hazards in Presence of Various Substances	Containers may explode in heat of fire. Do not cut, weld, heat, drill or pressurize empty container. Vapour explosion hazard indoors, outdoors or in sewers. Runoff to sewer may create fire or explosion hazard.
Products of Combustion	Carbon oxides (CO, CO ₂), nitrogen oxides (NO _x), sulphur oxides (SO _x), sulphur compounds (H ₂ S), water vapour (H ₂ O), smoke and irritating vapours as products of incomplete combustion.		

Fire Fighting Media and Instructions

NAERG96, GUIDE 128, Flammable liquids (Non-polar/Water-immiscible).

CAUTION: This product has a moderate flash point above 40°C: Use of water spray when fighting fire may be inefficient.

If tank, rail car or tank truck is involved in a fire, ISOLATE for 800 meters (1/2 mile) in all directions; also consider initial evacuation for 800 meters (1/2 mile) in all directions.

SMALL FIRES: Dry chemical, CO₂, water spray or regular foam.

LARGE FIRES: Water spray, fog or regular foam. Do not use straight streams. Move containers from fire area if you can do it without risk.

Fires Involving Tanks or Car/Trailer Loads: Fight fire from maximum distance or use unmanned hose holders or monitor nozzles.

Cool containers with flooding quantities of water until well after fire is out. Withdraw immediately in case of rising sound from venting devices or any discolouration of tank. ALWAYS stay away from the ends of tanks. For massive fire, use unmanned hose holders or monitor nozzles; if this is impossible withdraw from area and let fire burn. Wear positive pressure self-contained breathing apparatus (SCBA). Structural firefighters' protective clothing will only provide limited protection.

Section 6. Accidental Release Measures**Material Release or Spill**

NAERG96, GUIDE 128, Flammable Liquids (Non-polar/ Water-immiscible).

ELIMINATE ALL IGNITION SOURCES. Avoid contact. Stop leak if without risk. Contain spill. Absorb with inert absorbents, dry clay, or diatomaceous earth. Avoid inhaling dust of diatomaceous earth for it may contain silica in very fine particle size, making this a potential respiratory hazard. Place used absorbent in closed metal containers for later disposal or burn absorbent in a suitable combustion chamber. DO NOT FLUSH TO SEWERS, STREAMS OR OTHER BODIES OF WATER. Check with applicable jurisdiction for specific disposal requirements of spilled material and empty containers. Notify the appropriate authorities immediately.

Section 7. Handling and Storage**Handling**

Keep away from heat. Keep away from sources of ignition. Empty containers pose a fire risk. DO NOT reuse empty containers without commercial cleaning or reconditioning. Ground/bond line and equipment during pumping or transfer to avoid accumulation of static charge. DO NOT ingest. Do not breathe gas/vapour/spray. In case of insufficient ventilation, wear suitable respiratory equipment. If ingested, seek medical advice immediately. Avoid contact with skin and eyes. Practice good personal hygiene. Wash hands after handling and before eating. Launder work clothes frequently. Discard saturated leather goods.

Storage

Store in tightly closed containers in cool, dry, isolated, well-ventilated area, and away from incompatibles. Ground all equipment containing material.

Section 8. Exposure Controls/Personal Protection**Engineering Controls**

For normal application, special ventilation is not necessary. If user's operations generate vapours or mist, use ventilation to keep exposure to airborne contaminants below the exposure limit. Make-up air should always be supplied to balance air removed by exhaust ventilation. Ensure that eyewash station and safety shower are close to work-station.

Personal Protection -

The selection of personal protective equipment varies, depending upon conditions of use.

Eyes

Eye protection (i.e., safety glasses, safety goggles and/or face shield) should be determined based on conditions of use. If product is used in an application where splashing may occur, the use of safety goggles and/or a face shield should be considered.

Body

Wear appropriate clothing to prevent skin contact. As a minimum long sleeves and trousers should be worn.

Respiratory

Where concentrations in air may exceed the occupational exposure limits given in Section 2 (and those applicable to your area) and where engineering, work practices or other means of exposure reduction are not adequate, NIOSH approved respirators may be necessary to prevent overexposure by inhalation.

Hands

Wear appropriate chemically protective gloves. When handling hot product ensure gloves are heat resistant and insulated.

Feet

Wear appropriate footwear to prevent product from coming in contact with feet and skin.

Section 9. Physical and Chemical Properties

Physical State and Appearance	Bright oily liquid.	Viscosity	1.3-4.1 cSt @ 40°C (104°F)
Colour	Clear to yellow / brown. Low sulphur diesel fuels (<0.05 wt % sulphur) are colourless to light yellow (and may be dyed red for taxation purposes). Regular sulphur diesel fuels (0.05-0.50 % sulphur) may be colourless to yellow / brown and are usually dyed red for taxation purposes.	Pour Point	Variable, 0°C to -50°C (32°F to -58°F)
Odour	Petroleum oil like.	Softening Point	Not applicable.
Odour Threshold	Not available	Dropping Point	Not applicable.
Boiling Point	150-371°C (302-700°F)	Penetration	Not applicable.
Density	0.85 kg/L @ 15°C (Water = 1).	Oil / Water Dist. Coefficient	Not available
Vapour Density	4.5 (Air = 1)	Ionicity (in water)	Not applicable.

DIESEL FUEL		Page Number: 3	
Vapour Pressure	1.0 kPa @ 20°C (7.5 mmHg @ 68°F).	Dispersion Properties	Not available
Volatility	<0.1 (Butyl acetate = 1), less than gasoline.	Solubility	Insoluble in cold water, soluble in non-polar hydrocarbon solvents.

Section 10. Stability and Reactivity			
Corrosivity	Not available		
Stability	The product is stable under normal handling and storage conditions.	Hazardous Polymerization	Will not occur under normal working conditions.
Incompatible Substances / Conditions to Avoid	Reactive with oxidizing agents and acids.	Decomposition Products	May release COx, NOx, SOx, H2S, H2O, smoke and irritating vapours when heated to decomposition.

Section 11. Toxicological Information	
Routes of Entry	Skin contact, eye contact, inhalation, and ingestion.
Acute Lethality	Acute oral toxicity (LD50): 7500 mg/kg (rat).
Chronic or Other Toxic Effects	
Dermal Route:	Skin contact may cause moderate to severe irritation. Repeated exposure would produce drying and cracking or defatting dermatitis.
Inhalation Route:	Inhalation of vapours can cause CNS depression with symptoms of nausea, headaches, vomiting, dizziness, fatigue, light-headedness, reduced coordination, unconsciousness and possibly death. Inhalation can also cause irritation of nose and throat.
Oral Route:	Aspiration of liquid drops into the lungs may produce potentially fatal chemical pneumonitis (fluid in the lungs), severe lung damage, or respiratory failure.
Eye Irritation/Inflammation:	Eye contact may cause mild irritation, but no permanent damage.
Immunotoxicity:	Not available
Skin Sensitization:	This product is not expected to be a skin sensitizer, based on the available data and the known hazards of the components.
Respiratory Tract Sensitization:	This product is not expected to be a respiratory tract sensitizer, based on the available data and the known hazards of the components.
Mutagenic:	This product is not expected to be a mutagen, based on the available data and the known hazards of the components.
Reproductive Toxicity:	This product is not expected to be a reproductive hazard, based on the available data and the known hazards of the components.
Teratogenicity/Embryotoxicity:	This product is not expected to be a teratogen or an embryotoxin, based on the available data and the known hazards of the components.
Carcinogenicity (ACGIH):	ACGIH Notice of Intended Change (2000): proposed A3: animal carcinogen. [Diesel oil]
Carcinogenicity (IARC):	This product is not known to contain any chemicals at reportable quantities that are listed as group 1, 2A or 2B carcinogens by IARC.
Carcinogenicity (NTP):	This product is not known to contain any chemicals at reportable quantities that are listed as carcinogens by NTP.
Carcinogenicity (IRIS):	Not available
Carcinogenicity (OSHA):	This product is not known to contain any chemicals at reportable quantities that are listed as carcinogens by OSHA.
Other Considerations	No additional remark.

Section 12. Ecological Information			
Environmental Fate	Not available	Persistence/Bioaccumulation Potential	Not available
BOD5 and COD	Not available	Products of Biodegradation	Not available
Additional Remarks	No additional remark.		

Section 13. Disposal Considerations

Waste Disposal	Preferred waste management priorities are: (1) recycle or reprocess; (2) incineration with energy recovery; (3) disposal at licensed waste disposal facility. Ensure that disposal or reprocessing is in compliance with government requirements and local disposal regulations. Consult your local or regional authorities.
-----------------------	--

Section 14. Transport Information

TDG Classification	Diesel Fuel UN1202 3 III	Special Provisions for Transport	Not applicable.
---------------------------	-----------------------------------	---	-----------------

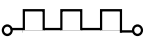
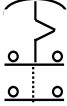
Section 15. Regulatory Information

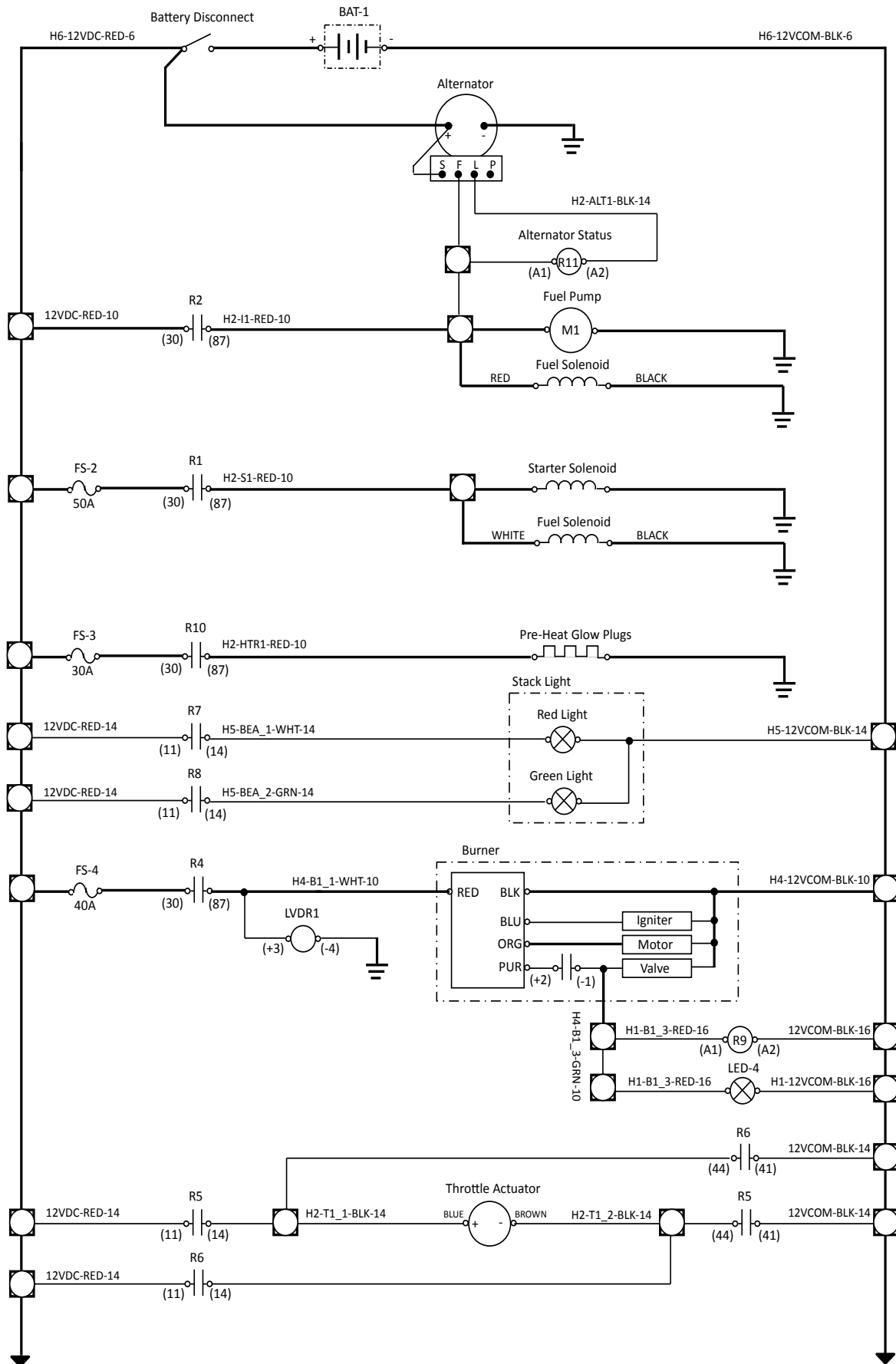
Other Regulations		This product is acceptable for use under the provisions of WHMIS-CPR. All components of this formulation are listed on the CEPA-DSL (Domestic Substances List).												
		All components of this formulation are listed on the US EPA-TSCA Inventory.												
		All components of this product are on the European Inventory of Existing Commercial Chemical Substances (EINECS).												
		This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all of the information required by the CPR.												
		Please contact Product Safety for more information.												
DSD/DPD (Europe)		Not evaluated.		HCS (U.S.A.)	CLASS: Irritating substance. CLASS: Target organ effects. CLASS: Combustible liquid having a flash point between 37.8°C (100°F) and 93.3°C (200°F).									
ADR (Europe) (Pictograms)		NOT EVALUATED FOR EUROPEAN TRANSPORT NON ÉVALUÉ POUR LE TRANSPORT EUROPÉEN.		DOT (U.S.A) (Pictograms)										
HMIS (U.S.A.)		<table><tr><td>Health Hazard</td><td>2*</td></tr><tr><td>Fire Hazard</td><td>2</td></tr><tr><td>Reactivity</td><td>0</td></tr><tr><td>Personal Protection</td><td>H</td></tr></table>		Health Hazard	2*	Fire Hazard	2	Reactivity	0	Personal Protection	H	NFPA (U.S.A.) Health  Fire Hazard Reactivity Specific hazard		Rating 0 Insignificant 1 Slight 2 Moderate 3 High 4 Extreme
Health Hazard	2*													
Fire Hazard	2													
Reactivity	0													
Personal Protection	H													

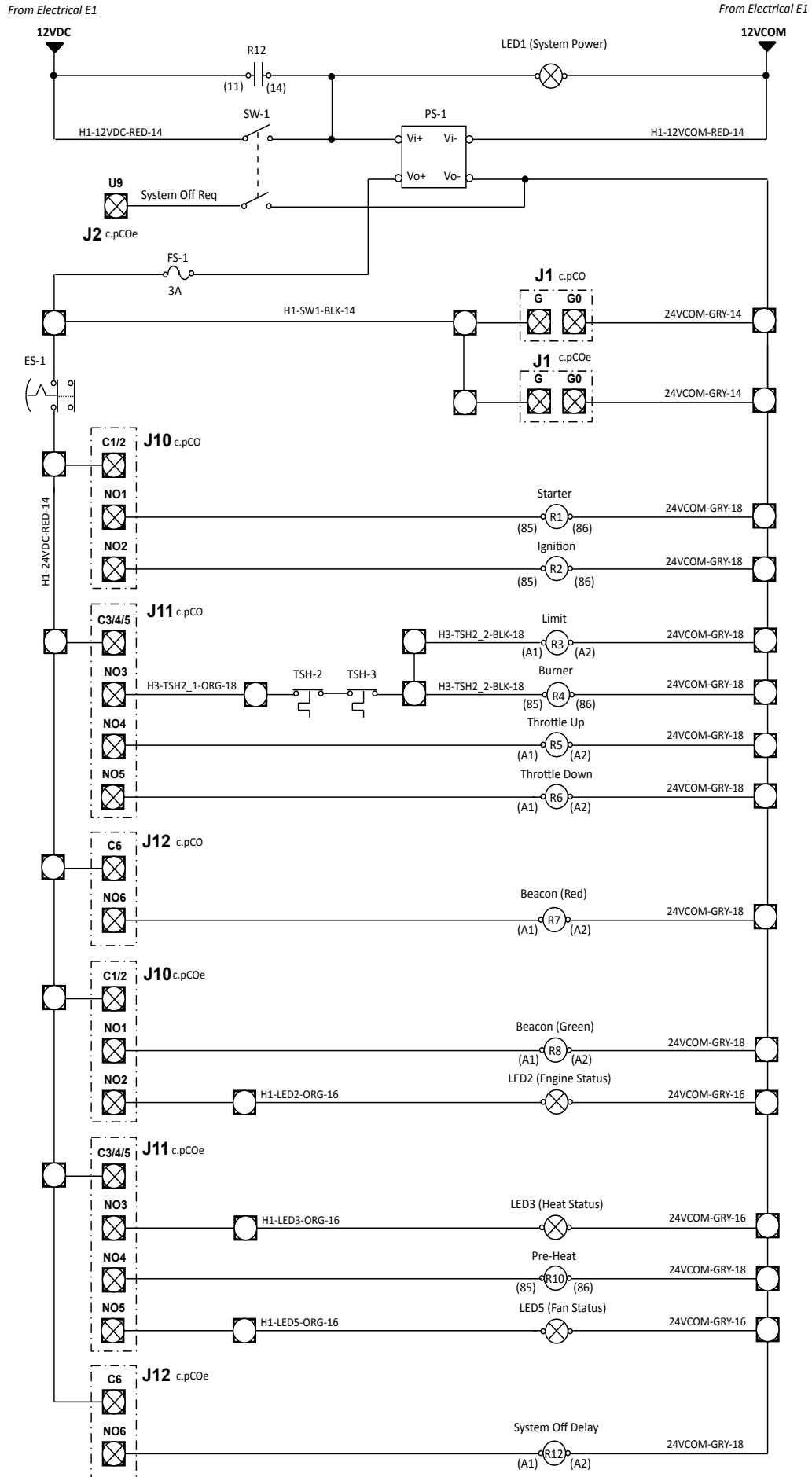
Section 16. Other Information

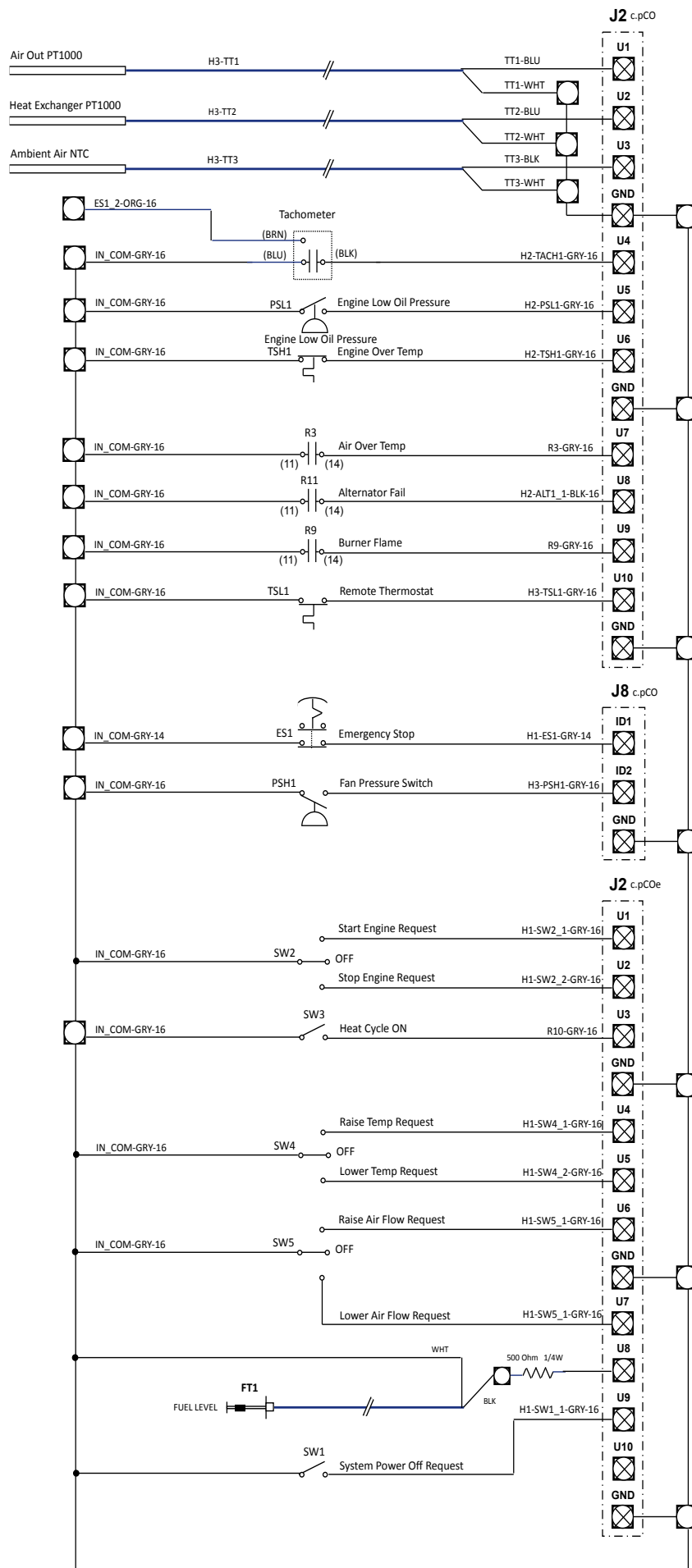
References	Available upon request. * Marque de commerce de Petro-Canada - Trademark		
Glossary	ACGIH - American Conference of Governmental Industrial Hygienists ADR - Agreement on Dangerous goods by Road (Europe) ASTM - American Society for Testing and Materials (BOD5 - Biological Oxygen Demand in 5 days CAN/CGA B149.2 Propane Installation Code CAS - Chemical Abstract Services CEPA - Canadian Environmental Protection Act CERCLA - Comprehensive Environmental Response, Compensation and Liability Act CFR - Code of Federal Regulations CHIP - Chemicals Hazard Information and Packaging Approved Supply List COD5 - Chemical Oxygen Demand in 5 days CPR - Controlled Products Regulations DOT - Department of Transport DSCL - Dangerous Substances Classification and Labeling (Europe) DSD/DPD - Dangerous Substances or Dangerous Preparations Directives (Europe) DSL - Domestic Substance List EEC/EU - European Economic Community/European Union EINECS - European Inventory of Existing Commercial Chemical Substances EPCRA - Emergency Planning and Community Right to Know Act FDA - Food and Drug Administration FIFRA - Federal Insecticide, Fungicide and Rodenticide Act HCS - Hazardous Communication System HMIS - Hazardous Material Information System IARC - International Agency for Research on Cancer IRIS - Integrated Risk Information System LD50/LC50 - Lethal Dose/Concentration kill 50% LDLo/LCLo - Lowest Published Lethal Dose/Concentration NAERG'96 - North American Emergency Response Guide Book (1996) NFPA - National Fire Prevention Association NIOSH - National Institute for Occupational Safety & Health NPRI - National Pollutant Release Inventory NSNR - New Substances Notification Regulations (Canada) NTP - National Toxicology Program OSHA - Occupational Safety & Health Administration PEL - Permissible Exposure Limit RCRA - Resource Conservation and Recovery Act SARA - Superfund Amendments and Reorganization Act SD - Single Dose STEL - Short Term Exposure Limit (15 minutes) TDG - Transportation Dangerous Goods (Canada) TDLo/TCLo - Lowest Published Toxic Dose/Concentration Tlm - Median Tolerance Limit TLV-TWA - Threshold Limit Value-Time Weighted Average TSCA - Toxic Substances Control Act USEPA - United States Environmental Protection Agency USP - United States Pharmacopoeia WHMIS - Workplace Hazardous Material Information System		
For Copy of MSDS	Prepared by Product Safety - TAR on 3/2/2001. Data entry by Product Safety - JDW.		
Fuels & Solvents:			
Western Canada, telephone: 403-296-4158; fax: 403-296-6551 Ontario & Central Canada, telephone: 1-800-668-0220; fax: 1-800-837-1228 Quebec & Eastern Canada, telephone: 514-640-8308; fax: 514-640-8385			
For Product Safety Information: (905) 804-4752			

Schematics

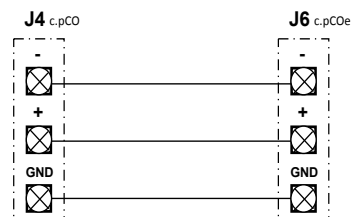
Schematic Legend		
Symbol	Device ID	Description
	FS-x	Fuse
	SW-x	Switch
	TB-x	PLC Terminal
	Rx	NO Relay Contact
	Rx	Relay Coil
	Sx	Solenoid
	Hx	Heater
	BAT-x	Battery
	Lx	Light
	TSH-x	High Temp SW
	TSL-x	Low Temp SW
	PSL-x	Low Pressure SW
	PSH-x	High Pressure SW
	ES-x	E-Stop
	TB-x	C/P Terminal



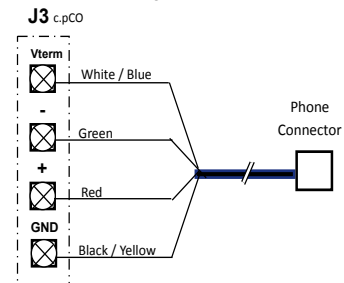




COMMUNICATION



DISPLAY





1205 SHERWIN RD
WINNIPEG, MANITOBA
CANADA R3H 0V3



WWW.HEAT-KING.CA



Support@tamarack-ind.com



1-8800-661-0304



1-204-885-7557



HEATKINGTECH